

//WallPaper prog

```
package com.example.myapplication;

import android.os.Bundle;
import android.app.WallpaperManager;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.drawable.AnimationDrawable;
import android.graphics.drawable.BitmapDrawable;
import android.graphics.drawable.Drawable;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import java.io.IOException;
import java.util.Timer;
import java.util.TimerTask;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    Button changewallpaper;
    Timer mytimer;
    Drawable drawable;
    WallpaperManager wpm;
    int id=1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        mytimer = new Timer ();
        wpm = WallpaperManager.getInstance(this);
        changewallpaper = findViewById(R.id.btn_click);
        changewallpaper.setOnClickListener(new View.OnClickListener()
        {
            @Override public void onClick(View view) {
                setWallpaper();
            }
        });
    }
    private void setWallpaper()
    {
        mytimer.schedule(new TimerTask()
        {
            @Override
            public void run() {
                if(id==1) {
                    drawable = getResources().getDrawable(R.drawable.one);
                    id = 2;
                }
                else if(id==2) {
                    drawable = getResources().getDrawable(R.drawable.two); id=3;
                }
                else if(id==3) {
                    drawable = getResources().getDrawable(R.drawable.three); id=4;
                }
                else if(id==4) {
                    drawable = getResources().getDrawable(R.drawable.four); id=5;
                }
                else if(id==5) {
```

```

        drawable = getResources().getDrawable(R.drawable.five); id=1;
    }
    Bitmap wallpaper = ((BitmapDrawable)drawable).getBitmap();
    try {
        wpm.setBitmap(wallpaper);
    }
    catch (IOException e)
    { e.printStackTrace();
    }
    }
    },0,30000);
}
}

```

//Counter prog

```

package com.example.counter;

import android.os.Bundle;
import android.os.Handler;
import android.os.Message;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    TextView ctr;
    Button start, stop;
    int counter=0;
    boolean running=false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        start=(Button) findViewById(R.id.btn_start);
        stop=(Button) findViewById(R.id.btn_stop);
        ctr=(TextView) findViewById(R.id.counter);
        start.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                counter=0;
                running=true;
                new MyCounter().start();
            }
        });
    }
}

```

```

        }
    });
    stop.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            running=false;
        }
    });
}

Handler handler=new Handler()
{
    public void handleMessage(Message m)
    {
        ctr.setText(String.valueOf(m.what));
    }
};

class MyCounter extends Thread
{
    public void run()
    {
        while(running)
        {
            counter++;
            handler.sendMessage(counter);
            try {
                Thread.sleep(1000);
            }
            catch (Exception e)
            { }
        }
    }
}
}

```

// Signin-signUp

Signup.JAVA

```

package com.example.signup;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.content.Intent;
import android.view.View;
import android.widget.Toast;
import java.util.regex.Pattern;

```

```

public class Signup extends AppCompatActivity {
    EditText username, pwd;
    Button signup;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        username = (EditText) findViewById(R.id.username);
        pwd = (EditText) findViewById(R.id.password);
        signup = (Button) findViewById(R.id.signup);
        signup.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String email = username.getText().toString();
                String userpwd = pwd.getText().toString();
                if (email.matches("")) {
                    Toast.makeText(Signup.this, "Username cannot be empty", Toast.LENGTH_LONG).show();
                    return;
                }
                if (!isValidPassword(userpwd)) {
                    Toast.makeText(Signup.this, "Invalid Password", Toast.LENGTH_LONG).show();
                    return;
                }
                Intent intent = new Intent(Signup.this, Signin.class);
                intent.putExtra("email", email);
                intent.putExtra("userpwd", userpwd);
                startActivity(intent);
            }
        });
    }
    Pattern lowercase = Pattern.compile("^[a-z].*$");
    Pattern uppercase = Pattern.compile("^[A-Z].*$");
    Pattern number = Pattern.compile("^[0-9].*$");
    Pattern special = Pattern.compile("^[@#$$%^&*(){}.,:/.]*$");

    private boolean isValidPassword(String userpwd) {
        if (userpwd.length() < 8) {
            return false;
        }
        if (!lowercase.matcher(userpwd).matches()) {
            return false;
        }
        if (!uppercase.matcher(userpwd).matches()) {
            return false;
        }
        if (!number.matcher(userpwd).matches()) {
            return false;
        }
        if (!special.matcher(userpwd).matches()) {
            return false;
        }
        return true;
    }
}

```

Signin.JAVA

```

package com.example.signup;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class Signin extends AppCompatActivity {
    EditText eusername, epwd;
    Button login;
    int count=0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_signin);
        eusername = (EditText) findViewById(R.id.username);
        epwd = (EditText) findViewById(R.id.password);
        login = (Button) findViewById(R.id.login);
        String regemail=getIntent().getStringExtra("email");
        String regpwd=getIntent().getStringExtra("userpwd");
        login.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String newemail=eusername.getText().toString();
                String newpwd=epwd.getText().toString();
                if(regemail.equals(newemail) && regpwd.equals(newpwd)){
                    Toast.makeText(Signin.this, "Login Successful", Toast.LENGTH_LONG).show();
                    Intent intent=new Intent(Signin.this, Welcome.class);
                    startActivity(intent);
                    eusername.setText("");
                    epwd.setText("");
                }
                else{
                    count++;
                    Toast.makeText(Signin.this, "Login Failed "+count, Toast.LENGTH_LONG).show();
                    if(count==3)
                    {
                        login.setEnabled(false);
                        Toast.makeText(Signin.this, "Login Failed "+count, Toast.LENGTH_LONG).show();
                    }
                }
            }
        });
    }
}

```

//MEDICINE REMINDER

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context=".MainActivity">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:layout_marginRight="10dp"
    android:orientation="horizontal">
```

```
<TextView
    android:layout_width="100dp"
    android:layout_height="match_parent"
    android:layout_weight="1"
    android:text="MEDICINE DATABASE"
    android:textAllCaps="false"
    android:textColor="#740D0101"
    android:textSize="20sp" />
```

```
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:layout_marginRight="10dp"
    android:orientation="horizontal">
```

```
<TextView
    android:layout_width="75dp"
    android:layout_height="48dp"
    android:layout_weight="1"
    android:text="Insert"
    android:textAllCaps="false"
    android:textColor="#740D0101"
    android:textSize="20sp" />
```

```
<Switch
    android:id="@+id/switch1"
    android:layout_width="wrap_content"
    android:layout_height="48dp"
    android:layout_weight="1" />
```

```
<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="48dp"
    android:layout_weight="1"
    android:text="Fetch"
    android:textSize="20sp" />
```

```
</LinearLayout>
```

```
<LinearLayout
```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:layout_marginRight="10dp"
        android:orientation="horizontal">

        <TextView
            android:id="@+id/medtext"
            android:layout_width="100dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Medicine Name"
            android:textAllCaps="false"
            android:textColor="#740D0101"
            android:textSize="24sp" />

        <EditText
            android:id="@+id/medicinename"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:ems="10"
            android:hint="Enter Medicine Name"
            android:textSize="20sp" />
    </LinearLayout>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:layout_marginRight="10dp"
        android:orientation="horizontal">
        <TextView
            android:id="@+id/textView3"
            android:layout_width="100dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Date"
            android:textAllCaps="false"
            android:textColor="#740D0101"
            android:textSize="24sp" />

        <EditText
            android:id="@+id/date"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:ems="10"
            android:inputType="date"
            android:hint="Enter Date DD/MM/YYYY"
            android:textSize="24sp" />
    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:layout_marginRight="10dp"
        android:orientation="horizontal">

```

```

<TextView
    android:layout_width="100dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="Time of the day"
    android:textAllCaps="false"
    android:textColor="#740D0101"
    android:textSize="20sp" />

<Spinner
    android:id="@+id/spinner"
    android:layout_width="24dp"
    android:layout_height="48dp"
    android:layout_weight="1"
    android:entries="@array/timeoftheday" />
</LinearLayout>

<Button
    android:id="@+id/insert"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="INSERT" />

<Button
    android:id="@+id/fetch"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="FETCH" />

```

</LinearLayout>

Project Tab->res->values->String.xml (Spinner values)

```

<resources>
    <string name="app_name">MedicineReminder</string>
    <string-array name="timeoftheday">
        <item> MORNING </item>
        <item> AFTERNOON </item>
        <item> EVENING </item>
        <item> NIGHT </item>
    </string-array>
</resources>

```

Mainactivity.java

```

package com.example.medicinereminder;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CompoundButton;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.Switch;
import android.widget.TextView;
import android.widget.Toast;

```



```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText medname,meddate;
    Button insert,fetch;
    Spinner day;
    Switch switch1;
    TextView medtxt;
    DatabaseConnection dbconnection;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        medname=(EditText)findViewById(R.id.medicinename);
        meddate=(EditText)findViewById(R.id.date);
        insert=(Button)findViewById(R.id.insert);
        fetch=(Button)findViewById(R.id.fetch);
        day=(Spinner)findViewById(R.id.spinner);
        switch1=(Switch)findViewById(R.id.switch1);
        medtxt=(TextView)findViewById(R.id.medtext);
        dbconnection=new DatabaseConnection(this);

        fetch.setVisibility(View.INVISIBLE);
        switch1.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(CompoundButton compoundButton, boolean b) {
                if(!b)
                {
                    fetch.setVisibility(View.INVISIBLE);
                    insert.setVisibility(View.VISIBLE);
                    medname.setVisibility(View.VISIBLE);
                    medtxt.setVisibility(View.VISIBLE);
                }
                else {
                    fetch.setVisibility(View.VISIBLE);
                    insert.setVisibility(View.INVISIBLE);
                    medname.setVisibility(View.INVISIBLE);
                    medtxt.setVisibility(View.INVISIBLE);
                }
            }
        });

        insert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String name=medname.getText().toString();
                String date=meddate.getText().toString();
                String time=day.getSelectedItem().toString();
                boolean insert=dbconnection.insertvalues(name,date,time);
                if(insert==true)
                {
                    Toast.makeText(getApplicationContext(), "Data Inserted", Toast.LENGTH_LONG).show();
                    medname.setText(" ");
                }
            }
        });
    }
}

```

```

        meddate.setText(" ");
    }
    else {
        Toast.makeText(getApplicationContext(), "Data Not Inserted", Toast.LENGTH_LONG).show();
    }
}
});

fetch.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        String date=meddate.getText().toString();
        String time=day.getSelectedItem().toString();
        String med="";
        Cursor c=dbconnection.FetchData(date,time);
        if(c.moveToFirst()) {
            do {
                med += (String.valueOf(c.getString(c.getColumnIndexOrThrow("MedicineName"))));
                med += "\n";
            } while (c.moveToNext());
            Toast.makeText(getApplicationContext(), med, Toast.LENGTH_LONG).show();
        }
        else
            Toast.makeText(getApplicationContext(), "No Entries in Database", Toast.LENGTH_SHORT).show();
    }
});
}

}

```

DatabaseConnection.java

```

package com.example.medicinereminder;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.annotation.Nullable;

public class DatabaseConnection extends SQLiteOpenHelper {
    public DatabaseConnection(Context context) {
        super(context, "Medicinedb", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        sqLiteDatabase.execSQL("create Table MDTable(MedicineName Text,date TEXT,time TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int i1) {
    }
}

```

```

public boolean insertvalues(String medname,String meddate,String medtime)
{
    SQLiteDatabase database=this.getWritableDatabase();
    ContentValues contentValues=new ContentValues();
    contentValues.put("MedicineName",medname);
    contentValues.put("date",meddate);
    contentValues.put("time",medtime);
    long res=database.insert("MDTable",null,contentValues);
    if(res==-1)
        return false;
    else
        return true;
}

public Cursor FetchData(String date,String time)
{
    SQLiteDatabase database=this.getReadableDatabase();
    Cursor c=database.rawQuery("Select * from MDTable where date='"+date+"' AND time='"+time+"' ",null);
    return c;
}
}

```